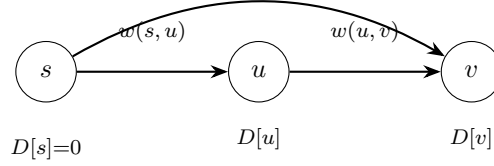


1. Checking Correctness

Algorithm



Return **true** iff all three conditions hold:

1. $D[s] = 0$
2. $\forall (u, v) \in E: D[v] \leq D[u] + w(u, v)$ (triangle inequality)
3. $\forall v \neq s \text{ with } D[v] < \infty: \exists (u, v) \in E \text{ with } D[v] = D[u] + w(u, v)$ (tightness)

Proof of correctness

Necessary: Shortest distances satisfy (1)–(3) by definition.

Sufficient ($D[v] = \text{dist}(s, v)$ **when** (1)–(3) **hold**):

Upper bound: For any path $s = u_0, u_1, \dots, u_k = v$, applying (2) inductively:

$$D[v] \leq D[u_{k-1}] + w(u_{k-1}, v) \leq \dots \leq \sum_{i=0}^{k-1} w(u_i, u_{i+1})$$

$$\implies D[v] \leq \text{dist}(s, v).$$

Lower bound: For v with $D[v] < \infty$, condition (3) gives a predecessor $\pi(v)$ with $D[v] = D[\pi(v)] + w(\pi(v), v)$. Since $w \geq 0$, $D[\pi(v)] \leq D[v]$, so the chain $v, \pi(v), \pi^2(v), \dots$ has non-increasing D -values and visits at most n distinct vertices before reaching s (where $D[s] = 0$). This chain yields a path from s to v of weight $D[v]$, so $\text{dist}(s, v) \leq D[v]$.

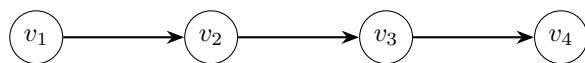
Unreachable vertices: If $D[v] = \infty$ but v is reachable, the upper bound gives $D[v] \leq \text{dist}(s, v) < \infty$, a contradiction. So $D[v] = \infty \iff v$ is unreachable. $\square$

Runtime: Conditions (1)–(3) each require a single scan of vertices/edges.

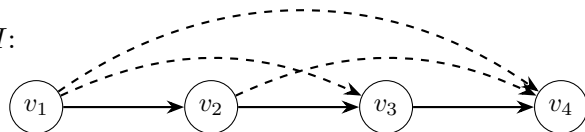
$$\boxed{O(m + n)}$$

2. Transitive Closure

G :



H :



begin

```

  TRANSITIVECLOSURE( $G = (V, E)$ )
  Compute a topological ordering  $v_1, v_2, \dots, v_n$  of  $G$ 
  for  $i \leftarrow n$  down to 1 do
    Reach[ $v_i$ ]  $\leftarrow$   $n$ -bit vector with bit  $v_i$  set
    for each  $(v_i, v_j) \in E$  do
      Reach[ $v_i$ ]  $\leftarrow$  Reach[ $v_i$ ]  $\mid$  Reach[ $v_j$ ]
  return  $H = \{(u, v) : \text{bit } v \text{ is set in Reach}[u]\}$ 

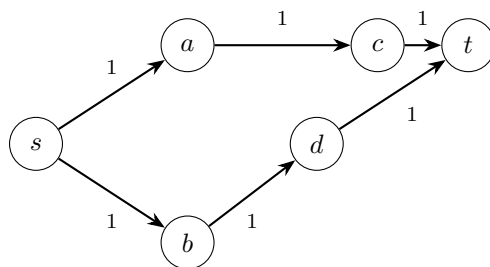
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Correctness: Processing in reverse topological order ensures $\text{Reach}[v_j]$ is complete before any predecessor v_i reads it (since $i < j$ for every edge (v_i, v_j)). After processing v_i , $\text{Reach}[v_i]$ contains all vertices reachable from v_i .

Runtime: Topological sort $O(m + n)$. Each of m edges triggers one OR of n -bit vectors ($O(n)$ each).

$$\boxed{O(mn)}$$

3. Fault-Tolerant Paths



All capacities = 1; two edge-disjoint s - t paths

Solution 1:

Reduction: Assign capacity 1 to every edge. By Menger's theorem, the maximum number of edge-disjoint s - t paths equals the max flow from s to t in this unit-capacity network.

1. Build unit-capacity network on G .
2. Run Edmonds-Karp (BFS-based Ford-Fulkerson). Each augmenting path finds one unit of flow.
3. If max flow ≥ 2 : decompose the flow into 2 paths by tracing from s along edges with flow = 1 to t , removing used edges after each trace.
4. If max flow < 2 : report no two edge-disjoint s - t paths exist.

Runtime: ≤ 2 BFS iterations, each $O(m)$. Flow decomposition: $O(m)$. $O(m)$

Solution 2:

Reduction: Assign capacity 1 to every edge.

1. Build unit-capacity network on G . Initialize flow $f(e) = 0$ for all e .
2. **Repeat up to 2 times:** Run DFS on the residual graph to find an augmenting s - t path. If found, push 1 unit of flow along the path and update the residual graph. If no path exists, stop.
3. If total flow ≥ 2 : decompose into 2 edge-disjoint paths. Else report no two edge-disjoint s - t paths exist.

Decomposition: Trace from s along edges with $f(e) = 1$ until t ; remove used edges; repeat.

Correctness:

$(\Rightarrow) \exists$ two edge-disjoint s - t paths $\Rightarrow$ send 1 unit along each $\Rightarrow$ feasible flow of value 2 $\Rightarrow$ max flow ≥ 2 .

$(\Leftarrow)$ Max flow $\geq 2 \Rightarrow \exists$ integer flow with $\forall e: f(e) \in \{0, 1\}$ (integrality theorem). Flow conservation $\Rightarrow$ tracing from s along $f=1$ edges reaches t . Remove first path; remaining flow has value 1; repeat $\Rightarrow$ two edge-disjoint paths.

Runtime: ≤ 2 DFS iterations, each $O(m + n)$. Decomposition: $O(m)$. $\boxed{O(m + n)}$

(a)

Yes. The same reduction generalizes: k edge-disjoint s - t paths exist $\iff$ max flow $\geq k$ in the unit-capacity network. Run Ford-Fulkerson for up to k augmenting path iterations.

(b)

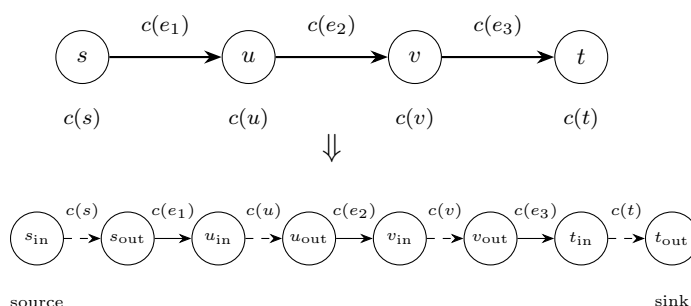
k DFS augmenting path iterations $\times O(m + n)$ each $+ O(km)$ decomposition.

$$\boxed{O(k(m + n))}$$

4. Vertex Capacities

Reduction: Split each vertex v into v_{in} and v_{out} :

- All incoming edges of v redirect to v_{in}
- All outgoing edges of v redirect from v_{out}
- Add internal edge $(v_{\text{in}}, v_{\text{out}})$ with capacity $c(v)$
- Original edge (u, v) with capacity $c(e)$ becomes $(u_{\text{out}}, v_{\text{in}})$ with capacity $c(e)$



New source = s_{in} , new sink = t_{out} . The new graph G' has $2n$ vertices and $m + n$ edges.

Correctness:

$(\Rightarrow)$ Given a feasible flow f' in G' , define $f(u, v) = f'(u_{\text{out}}, v_{\text{in}})$ in G . Edge capacities are preserved. The total flow through v equals $f'(v_{\text{in}}, v_{\text{out}}) \leq c(v)$, so the vertex capacity is satisfied.

($\Leftarrow$) Given a feasible flow f in G with vertex capacities, define:

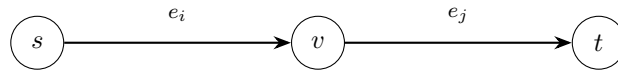
$$f'(u_{\text{out}}, v_{\text{in}}) = f(u, v), \quad f'(v_{\text{in}}, v_{\text{out}}) = \sum_u f(u, v)$$

Since f respects $c(v)$: $f'(v_{\text{in}}, v_{\text{out}}) \leq c(v)$. Conservation in G' follows from conservation in G .

$\Rightarrow$ Max flow in $G' = \text{max flow in } G \text{ with vertex capacities. Run any max-flow algorithm on } G'.$ $\square$

Reduce to standard max-flow on G' with $2n$ vertices, $m + n$ edges

5. Convex Combination of Flows



$$A[i], B[i] \leq \text{cap}(e_i)$$

$$A[j], B[j] \leq \text{cap}(e_j)$$

Let f_1, f_2 be feasible flows with arrays $A[1..m], B[1..m]$. Define $C[i] = c \cdot A[i] + (1 - c) \cdot B[i]$ for $0 \leq c \leq 1$.

Capacity constraints: Since $c \geq 0, 1 - c \geq 0, A[i] \geq 0, B[i] \geq 0$:

$$C[i] = c \cdot A[i] + (1 - c) \cdot B[i] \geq 0$$

Since $A[i] \leq \text{cap}(e_i)$ and $B[i] \leq \text{cap}(e_i)$:

$$C[i] \leq c \cdot \text{cap}(e_i) + (1 - c) \cdot \text{cap}(e_i) = \text{cap}(e_i)$$

Flow conservation: For each vertex $v \neq s, t$:

$$\begin{aligned} \sum_{\text{in}} C[i] - \sum_{\text{out}} C[i] &= c \left(\sum_{\text{in}} A[i] - \sum_{\text{out}} A[i] \right) + (1 - c) \left(\sum_{\text{in}} B[i] - \sum_{\text{out}} B[i] \right) \\ &= c \cdot 0 + (1 - c) \cdot 0 = 0 \end{aligned}$$

$\Rightarrow C[1..m]$ satisfies capacity constraints and flow conservation $\Rightarrow C$ is a feasible flow. $\square$

$$|f_C| = c \cdot |f_1| + (1 - c) \cdot |f_2|$$